Dietary flaxseed lignan extract lowers plasma cholesterol and glucose concentrations in hypercholesterolaemic subjects.

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Lignans, derived from flaxseed, are phyto-oestrogens being increasingly studied for their health benefits. An 8-week, randomised, double-blind, placebo-controlled study was conducted in fifty-five hypercholesterolaemic subjects, using treatments of 0 (placebo), 300 or 600 mg/d of dietary secoisolariciresinol diglucoside (SDG) from flaxseed extract to determine the effect on plasma lipids and fasting glucose levels. Significant treatment effects were achieved ($P < 0.05$ to < 0.001) for the decrease of total cholesterol (TC), LDL-cholesterol (LDL-C) and glucose concentrations, as well as their percentage decrease from baseline. At weeks 6 and 8 in the 600 mg SDG group, the decreases of TC and LDL-C concentrations were in the range from 22.0 to 24.38 % respectively (all $P < 0.005$ compared with placebo). For the 300 mg SDG group, only significant differences from baseline were observed for decreases of TC and LDL-C. A substantial effect on lowering concentrations of fasting plasma glucose was also noted in the 600 mg SDG group at weeks 6 and 8, especially in the subjects with baseline glucose concentrations > or = 5.83 mmol/l (lowered 25.56 and 24.96 %; $P = 0.015$ and $P = 0.012$ compared with placebo, respectively). Plasma concentrations of secoisolariciresinol (SECO), enterodiol (ED) and enterolactone were all significantly raised in the groups supplemented with flaxseed lignan. The observed cholesterol-lowering values were correlated with the concentrations of plasma SECO and ED (r 0.128-0.302; $P < 0.05$ to < 0.001). In conclusion, dietary flaxseed lignan extract decreased plasma cholesterol and glucose concentrations in a dose-dependent manner.